

DATABASE DESIGN DOCUMENT

Project Title: Thrift Store Management System

WEEK 2: ENTITY, ATTRIBUTES AND ENTITY RELATIONSHIP DIAGRAM

✓ Entity Identification

1. User

Primary Key (PK): User_ID

Attributes:

- User_ID
 - User_Name
 - Phone_Number
 - Email
 - Address
 - Password
 - Role (Customer / Seller)
-

2. Product

Primary Key (PK): Product_ID

Foreign Key (FK):

- User_ID

Attributes:

- Product_ID
- Product_Name
- Description
- Category
- Condition
- Price

- Product_Image
 - Availability_Status
-

3. Category

Primary Key (PK): Category_ID

Attributes:

- Category_ID
 - Category_Name
 - Category_Description
-

4. Order

Primary Key (PK): Order_ID

Foreign Keys (FK):

- User_ID

Attributes:

- Order_ID
 - Order_Date
 - Total_Amount
 - Order_Status
-

5. Order Item

Primary Key (PK): OrderItem_ID

Foreign Keys (FK):

- Order_ID
- Product_ID

Attributes:

- OrderItem_ID
- Quantity

- Price
-

6. Payment

Primary Key (PK): Payment_ID

Foreign Key (FK):

- Order_ID

Attributes:

- Payment_ID
 - Payment_Method
 - Payment_Date
 - Amount
 - Payment_Status
 - Transaction_ID
-

7. Order Tracking

Primary Key (PK): Tracking_ID

Foreign Key (FK):

- Order_ID

Attributes:

- Tracking_ID
 - Tracking_Status
 - Updated_Date
 - Delivery_Date
-

8. Admin

Primary Key (PK): Admin_ID

Attributes:

- Admin_ID

- Admin_Name
 - Username
 - Password
 - Email
-

✓ Relationships

Relationship	Cardinality
User → Product	One User (Seller) can list Many Products (1:M)
Category → Product	One Category contains Many Products (1:M)
User → Order	One User can place Many Orders (1:M)
Order → Order Item	One Order contains Many Order Items (1:M)
Product → Order Item	One Product can appear in Many Order Items (1:M)
Order → Payment	One Order has One Payment (1:1)
Order → Order Tracking	One Order has One Tracking Record (1:1)
Admin → Product	One Admin manages Many Products (1:M)
Admin → Order	One Admin manages Many Orders (1:M)

✓ Entity–Relationship Model

Entities

- User
- Product
- Category
- Order
- Order Item
- Payment
- Order Tracking
- Admin

Primary Keys

- User_ID
- Product_ID
- Category_ID
- Order_ID
- OrderItem_ID
- Payment_ID
- Tracking_ID
- Admin_ID

Foreign Keys

- Product → User_ID
- Order → User_ID
- Order Item → Order_ID
- Order Item → Product_ID
- Payment → Order_ID
- Order Tracking → Order_ID

✓ Database Description

The **Thrift Store Management System** is designed to simplify the buying and selling of second-hand products through an online platform. Customers can register, log in, browse products by category, add products to their cart, place orders, complete secure payments, and track their orders. Sellers can upload, update, and manage product listings, while administrators oversee users, products, orders, and payments. The database securely stores customer, product, order, payment, and tracking information while maintaining relationships between entities using primary and foreign keys. This design minimizes data redundancy, improves data integrity, and provides an efficient solution for managing online thrift store operations.

ER DIAGRAM – THRIFT STORE MANAGEMENT SYSTEM

